

Constraint Optimization: Practice Set

Lagrangian, Duality, and Minimax

1 Solving Primal Problems (Lagrangian)

General Strategy

How to solve Primal Problems:

1. **Standard Form:** Ensure constraints are in the form $g(x) \leq 0$ or $h(x) = 0$.
2. **Lagrangian Construction:** Write $\mathcal{L}(x, \lambda) = f(x) + \sum \lambda_i g_i(x)$.
3. **KKT Conditions (The System to Solve):**
 - **Stationarity:** $\nabla_x \mathcal{L} = 0$ (Derivatives w.r.t variables must be 0).
 - **Primal Feasibility:** The original constraints must hold.
 - **Dual Feasibility:** $\lambda \geq 0$ (for inequality constraints).
 - **Complementary Slackness:** $\lambda \cdot g(x) = 0$ (Either $\lambda = 0$ or the constraint is active).
4. **Solve:** Use substitution to solve the system of equations for x^* .

Problem 1 (Easy - Weighted Quadratic):

Minimize $f(x, y) = 2x^2 + y^2$ subject to the equality constraint $x + y = 3$.

Step-by-Step Solution:

Step 1: Lagrangian Formulation: Since this is an equality constraint, we use a multiplier ν (unrestricted sign).

$$\mathcal{L}(x, y, \nu) = 2x^2 + y^2 + \nu(3 - x - y)$$

Step 2: Stationarity (Gradient = 0):

$$\frac{\partial \mathcal{L}}{\partial x} = 4x - \nu = 0 \implies x = \frac{\nu}{4} \quad \text{--- (Eq 1)}$$

$$\frac{\partial \mathcal{L}}{\partial y} = 2y - \nu = 0 \implies y = \frac{\nu}{2} \quad \text{--- (Eq 2)}$$

Step 3: Feasibility (Constraint): Substitute (Eq 1) and (Eq 2) into $x + y = 3$:

$$\frac{\nu}{4} + \frac{\nu}{2} = 3$$

$$\frac{\nu + 2\nu}{4} = 3 \implies 3\nu = 12 \implies \nu = 4$$

Step 4: Final Calculation: Substitute $\nu = 4$ back into Eq 1 and Eq 2:

$$x^* = \frac{4}{4} = 1, \quad y^* = \frac{4}{2} = 2$$

Optimal Value: $f(1, 2) = 2(1)^2 + (2)^2 = 6$.

Problem 2 (Medium - Linear Objective on Circle):Minimize $f(x, y) = -x - y$ subject to $x^2 + y^2 \leq 2$.**Step-by-Step Solution:****Step 1: Lagrangian Formulation:** Constraint: $x^2 + y^2 - 2 \leq 0$.

$$\mathcal{L} = -x - y + \lambda(x^2 + y^2 - 2)$$

Step 2: Stationarity:

$$\partial_x : -1 + 2\lambda x = 0 \implies x = \frac{1}{2\lambda}$$

$$\partial_y : -1 + 2\lambda y = 0 \implies y = \frac{1}{2\lambda}$$

Step 3: Slackness & Logic: From $\lambda(x^2 + y^2 - 2) = 0$: Either $\lambda = 0$ or $x^2 + y^2 = 2$. If $\lambda = 0$, the stationarity equations become $-1 = 0$, which is impossible. Therefore, $\lambda \neq 0$, so the constraint must be active ($x^2 + y^2 = 2$).

Step 4: Solving for λ :

$$\left(\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 = 2 \implies \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} = 2$$

$$\frac{2}{4\lambda^2} = 2 \implies 4\lambda^2 = 1 \implies \lambda^2 = \frac{1}{4} \implies \lambda = \frac{1}{2}$$

(We take positive λ for $\leq$ constraints).**Step 5: Final Calculation:**

$$x^* = \frac{1}{2(0.5)} = 1, \quad y^* = \frac{1}{2(0.5)} = 1$$

Optimal Value: $-1 - 1 = -2$.**Problem 3 (Hard - 3 Variables / Geometry):**Minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to the plane $x + 2y + 3z = 14$.**Step-by-Step Solution:****Step 1: Lagrangian Formulation:**

$$\mathcal{L} = x^2 + y^2 + z^2 + \nu(14 - x - 2y - 3z)$$

Step 2: Stationarity:

$$\partial_x : 2x - \nu = 0 \implies x = \nu/2$$

$$\partial_y : 2y - 2\nu = 0 \implies y = \nu$$

$$\partial_z : 2z - 3\nu = 0 \implies z = 3\nu/2$$

Step 3: Substitution: Substitute all into $x + 2y + 3z = 14$:

$$\left(\frac{\nu}{2}\right) + 2(\nu) + 3\left(\frac{3\nu}{2}\right) = 14$$

Multiply entire equation by 2 to clear fractions:

$$\nu + 4\nu + 9\nu = 28 \implies 14\nu = 28 \implies \nu = 2$$

Step 4: Final Result:

$$x^* = 1, \quad y^* = 2, \quad z^* = 3$$

Optimal Value: $1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$.

2 Converting Primal to Dual Formulation

General Strategy

How to Formulate the Dual: The goal is to find the function $g(\lambda)$ which depends *only* on the Lagrange multipliers.

1. **Write Lagrangian:** $\mathcal{L}(x, \lambda)$.
2. **Minimize $\mathcal{L}$ w.r.t x :** Treat λ as a constant number. Find the x values that minimize $\mathcal{L}$ (by setting derivatives to 0). Let's call these $x(\lambda)$.
3. **Substitute Back:** Plug $x(\lambda)$ back into the Lagrangian equation.
4. **Simplify:** The result is $g(\lambda)$. The Dual Problem is "Maximize $g(\lambda)$ subject to $\lambda \geq 0$ ".

Problem 4 (Easy - Shifted Quadratic):

Convert to Dual: Minimize $(x - 3)^2$ subject to $x \leq 1$.

Solution:

Step 1: Lagrangian: $\mathcal{L} = (x - 3)^2 + \lambda(x - 1)$.

Step 2: Minimize over x :

$$2(x - 3) + \lambda = 0 \implies 2x - 6 = -\lambda \implies x = 3 - \frac{\lambda}{2}$$

Step 3: Substitute x back: Term 1: $(x - 3)^2 = (3 - \frac{\lambda}{2} - 3)^2 = (-\frac{\lambda}{2})^2 = \frac{\lambda^2}{4}$. Term 2: $\lambda(x - 1) = \lambda(3 - \frac{\lambda}{2} - 1) = \lambda(2 - \frac{\lambda}{2}) = 2\lambda - \frac{\lambda^2}{2}$.

Step 4: Combine:

$$g(\lambda) = \frac{\lambda^2}{4} + 2\lambda - \frac{\lambda^2}{2} = 2\lambda - \frac{\lambda^2}{4}$$

Step 5: Dual Formulation: Maximize $2\lambda - \frac{\lambda^2}{4}$ subject to $\lambda \geq 0$.

Problem 5 (Medium - Entropy/Log):

Convert to Dual: Minimize $x \ln x + y \ln y$ subject to $x + y = 1$ ($x, y > 0$).

Solution:

Step 1: Lagrangian: $\mathcal{L} = x \ln x + y \ln y + \nu(1 - x - y)$.

Step 2: Minimize over x, y :

$$\partial_x : (1 \cdot \ln x + x \cdot \frac{1}{x}) - \nu = 0 \implies \ln x + 1 - \nu = 0 \implies \ln x = \nu - 1 \implies x = e^{\nu-1}$$

By symmetry, $y = e^{\nu-1}$.

Step 3: Substitute back:

$$\begin{aligned} g(\nu) &= e^{\nu-1}(\nu-1) + e^{\nu-1}(\nu-1) + \nu(1 - e^{\nu-1} - e^{\nu-1}) \\ &= 2(\nu-1)e^{\nu-1} + \nu - 2\nu e^{\nu-1} \\ &= 2\nu e^{\nu-1} - 2e^{\nu-1} + \nu - 2\nu e^{\nu-1} \end{aligned}$$

Step 4: Simplify: The $2\nu e^{\nu-1}$ terms cancel.

$$g(\nu) = \nu - 2e^{\nu-1}$$

Step 5: Dual Formulation: Maximize $\nu - 2e^{\nu-1}$.

Problem 6 (Hard - Multiple Constraints):

Convert to Dual: Minimize $\frac{1}{2}x^2 + \frac{1}{2}y^2$ subject to $x \geq 1$ and $y \geq 2$.

Solution:

Step 1: Standard Constraints: $1 - x \leq 0$ and $2 - y \leq 0$.

Step 2: Lagrangian: $\mathcal{L} = \frac{1}{2}x^2 + \frac{1}{2}y^2 + \lambda_1(1 - x) + \lambda_2(2 - y)$.

Step 3: Minimize over x, y :

$$\partial_x : x - \lambda_1 = 0 \implies x = \lambda_1$$

$$\partial_y : y - \lambda_2 = 0 \implies y = \lambda_2$$

Step 4: Substitute back:

$$\begin{aligned} g(\lambda) &= \left(\frac{1}{2}\lambda_1^2 + \lambda_1(1 - \lambda_1)\right) + \left(\frac{1}{2}\lambda_2^2 + \lambda_2(2 - \lambda_2)\right) \\ &= \left(\lambda_1 - \frac{1}{2}\lambda_1^2\right) + \left(2\lambda_2 - \frac{1}{2}\lambda_2^2\right) \end{aligned}$$

Step 5: Dual Formulation: Maximize $(\lambda_1 - 0.5\lambda_1^2) + (2\lambda_2 - 0.5\lambda_2^2)$ subject to $\lambda_1, \lambda_2 \geq 0$.

3 Solving Dual Problems

General Strategy

How to Solve Dual Problems: Here we perform the full cycle.

1. **Derive $g(\lambda)$:** (As done in Section 2).
2. **Maximize $g(\lambda)$:** Differentiate $g(\lambda)$ with respect to λ , set to 0, and solve for λ^* .
3. **Check Feasibility:** Ensure $\lambda^* \geq 0$. If the calculated λ is negative, the optimum is usually at $\lambda = 0$.
4. **Calculate Value:** Plug λ^* into $g(\lambda)$.

Problem 7 (Easy): Solve the Dual for: Minimize $f = x^2 + y^2$ subject to $2x + y \leq -5$.
(Note constraint: $2x + y + 5 \leq 0$).

Solution:

Step 1: Formulate Lagrangian:

$$\mathcal{L} = x^2 + y^2 + \lambda(2x + y + 5)$$

Step 2: Find Dual Function $g(\lambda)$: Min w.r.t x : $2x + 2\lambda = 0 \implies x = -\lambda$. Min w.r.t y :
 $2y + \lambda = 0 \implies y = -\lambda/2$. Substitute:

$$g(\lambda) = (-\lambda)^2 + \left(-\frac{\lambda}{2}\right)^2 + \lambda(2(-\lambda) + \left(-\frac{\lambda}{2}\right) + 5)$$

$$g(\lambda) = \lambda^2 + 0.25\lambda^2 - 2\lambda^2 - 0.5\lambda^2 + 5\lambda$$

$$g(\lambda) = 5\lambda - 1.25\lambda^2$$

Step 3: Maximize Dual:

$$g'(\lambda) = 5 - 2.5\lambda = 0 \implies 2.5\lambda = 5 \implies \lambda^* = 2$$

Step 4: Check: $\lambda^* = 2 \geq 0$. Valid.

Step 5: Final Dual Value:

$$g(2) = 5(2) - 1.25(2^2) = 10 - 5 = 5$$

Problem 8 (Medium): Solve the Dual for: Minimize $f = \frac{1}{2}x^2 + y$ subject to $y \geq 1$ and $x \geq 0$.

Solution:

Step 1: Primal Setup: Constraints are $-x \leq 0$ and $1 - y \leq 0$.

Step 2: Formulate Lagrangian:

$$\mathcal{L} = \frac{1}{2}x^2 + y + \lambda_1(-x) + \lambda_2(1 - y)$$

Step 3: Minimize w.r.t Primal variables (x, y) :

- $\partial_x \mathcal{L} = x - \lambda_1 = 0 \implies x^* = \lambda_1$.
- $\partial_y \mathcal{L} = 1 - \lambda_2 = 0$. (This implies λ_2 *must* be 1 for a finite solution, otherwise $\mathcal{L} \rightarrow -\infty$).

Step 4: Find Dual Function $g(\lambda_1, \lambda_2)$: Substitute $x = \lambda_1$:

$$g(\lambda) = \left(\frac{1}{2}\lambda_1^2 - \lambda_1^2\right) + y(1 - \lambda_2) + \lambda_2$$

Since $1 - \lambda_2 = 0$, the middle term vanishes.

$$g(\lambda_1, \lambda_2) = -\frac{1}{2}\lambda_1^2 + \lambda_2$$

Constraint on dual: $\lambda_2 = 1$ is required.

Step 5: Solve Dual Problem: Maximize $g = -\frac{1}{2}\lambda_1^2 + 1$ subject to $\lambda_1 \geq 0$. To maximize this, we choose the smallest λ_1 possible (since $-\lambda_1^2$ reduces the value). $\lambda_1^* = 0$.

Step 6: Final Value: $g(0, 1) = 0 + 1 = 1$.

Problem 9 (Hard): Solve the Dual for: Minimize $x^2 + y^2 + z^2$ subject to $x + y + z = 6$.

Solution:

Step 1: Lagrangian: $\mathcal{L} = x^2 + y^2 + z^2 + \nu(6 - x - y - z)$.

Step 2: Stationarity: $2x - \nu = 0 \implies x = \nu/2$. $2y - \nu = 0 \implies y = \nu/2$. $2z - \nu = 0 \implies z = \nu/2$.

Step 3: Dual Function $g(\nu)$: Sum of squares: $3(\nu/2)^2 = \frac{3\nu^2}{4}$. Constraint term: $\nu(6 - \frac{3\nu}{2}) = 6\nu - \frac{3\nu^2}{2}$.

$$g(\nu) = \frac{3\nu^2}{4} + 6\nu - \frac{6\nu^2}{4} = 6\nu - \frac{3\nu^2}{4}$$

Step 4: Maximize Dual:

$$g'(\nu) = 6 - \frac{6\nu}{4} = 6 - 1.5\nu = 0$$

$$1.5\nu = 6 \implies \nu = 4$$

Step 5: Final Value:

$$g(4) = 6(4) - \frac{3(16)}{4} = 24 - 12 = 12$$

4 Minimax Inequality

General Strategy

Concept: The Minimax inequality (Weak Duality) states that the Dual optimal (d^*) is a lower bound for the Primal optimal (p^*):

$$\max_{\lambda \geq 0} \min_x \mathcal{L}(x, \lambda) \leq \min_x \max_{\lambda \geq 0} \mathcal{L}(x, \lambda)$$

$$d^* \leq p^*$$

Strong Duality ($d^* = p^*$) holds for convex problems with feasible solutions (Slater's condition).

Problem 10 (Easy - Convex/Equality):

Verify Minimax for $f(x) = x^2$ s.t. $x \geq 2$.

Solution:

1. **Primal (RHS):** We want $\min x^2$ where $x \in [2, \infty)$. Obviously, min is at $x = 2$, value **4**. So $p^* = 4$.
2. **Dual (LHS):** From Problem 4 logic (shifted), $g(\lambda) = 2\lambda - \lambda^2/4$. Max of this occurs at $\lambda = 4$. Value: $2(4) - 16/4 = 8 - 4 = \mathbf{4}$. So $d^* = 4$.
3. **Conclusion:** $4 \leq 4$. (Equality holds).

Problem 11 (Medium - Non-Convex/Gap):

Consider $f(x) = x^3 - 3x$ subject to $x \geq -2$.

Solution:

1. **Primal (p^*):** Graph $x^3 - 3x$. Local min at 1 (val -2). Boundary at -2 (val -2). Global min is **-2**.
2. **Dual (d^*):** $\mathcal{L} = x^3 - 3x - \lambda(x + 2)$. As $x \rightarrow -\infty$, x^3 dominates and goes to $-\infty$. Therefore, $\min_x \mathcal{L} = -\infty$ for any fixed λ . The max of $-\infty$ is still $-\infty$.
3. **Conclusion:** $-\infty \leq -2$. (Weak duality holds, but there is a massive "Duality Gap" because the problem is not convex).

Problem 12 (Hard - Conceptual):

Why does Minimax hold?

Explanation:

1. For any feasible x (where $g(x) \leq 0$) and any valid $\lambda \geq 0$, the term $\lambda g(x)$ is ≤ 0 .
2. Therefore, $\mathcal{L}(x, \lambda) = f(x) + \lambda g(x) \leq f(x)$.

3. If we minimize the left side, it stays lower than the min of the right side.
4. The "Max-Min" is essentially trying to push the lower bound up as high as possible, but it can never cross the actual primal minimum.